

Attributes in DBMS

What is an Attribute?

- An attribute is a property or characteristic of an entity.
- Attributes describe entities in a database.
- In a table, attributes are represented as columns.

Example

Student Table

Student_ID	Name	Age	Course
101	Ravi	20	B.Tech

- Student_ID, Name, Age, and Course are attributes.

Important Points About Attributes

Basic Points

1. Attribute describes an entity.
2. Attributes are represented by columns in a table.
3. Each attribute has a name.
4. Each attribute stores specific information.
5. Attributes help identify records.
6. Attributes can store numbers, text, dates, etc.
7. Every entity has one or more attributes.
8. Attributes are important in database design.
9. Attribute values should be meaningful.
10. Attributes help organize data properly.

Types of Attributes in DBMS

1. Simple Attribute

11. Cannot be divided further.
12. Stores single value only.

Example

- Age
- Salary

2. Composite Attribute

- 13. Can be divided into smaller parts.
- 14. Contains multiple subparts.

Example

- Address → Street, City, State
- Name → First Name, Last Name

3. Single-Valued Attribute

- 15. Stores only one value for each entity.
- 16. One entity has only one value.

Example

- Date of Birth
- Roll Number

4. Multi-Valued Attribute

- 17. Stores multiple values for one entity.
- 18. One entity can have many values.

Example

- Phone Numbers
- Skills

5. Derived Attribute

- 19. Value can be calculated from another attribute.
- 20. Usually not stored directly.

Example

- Age derived from Date of Birth

6. Stored Attribute

- 21. Attribute stored physically in database.
- 22. Used to derive other attributes.

Example

- Date of Birth

7. Key Attribute

- 23. Used to uniquely identify records.
- 24. No two rows have same key value.

Example

- Student_ID
- Employee_ID

8. Null Attribute

- 25. Contains no value or unknown value.
- 26. Represents missing information.

Example

- Middle Name may be NULL

Additional Student Understanding Points

- 27. Attributes are shown as ovals in ER diagrams.
- 28. Key attributes are underlined in ER diagrams.
- 29. Multi-valued attributes are shown using double ovals.
- 30. Derived attributes are shown using dotted ovals.
- 31. Composite attributes contain sub-attributes.

- 32.Attributes improve data organization.
- 33.Correct attributes improve database efficiency.
- 34.Too many unnecessary attributes waste storage.
- 35.Attribute names should be simple.
- 36.Duplicate attributes should be avoided.
- 37.Attributes must store accurate data.
- 38.Data type is assigned to each attribute.
- 39.Example data types: INTEGER, VARCHAR, DATE.
- 40.Attributes help in searching data.
- 41.Attributes help in sorting data.
- 42.Attributes help in filtering records.
- 43.Attributes are used in queries.
- 44.Attributes help create relationships between tables.
- 45.Primary keys are important attributes.
- 46.Foreign keys connect tables.
- 47.Attributes should follow normalization rules.
- 48.Proper attributes reduce redundancy.
- 49.Attribute values should follow constraints.
- 50.Attributes are the basic building blocks of a database.

Example of Attributes in Student Database

Attribute Name	Description
Student_ID	Unique student number
Name	Student name
Age	Student age
Gender	Student gender
Phone_Number	Contact number
Address	Student address

Attribute Name	Description
Course	Course name

Short Definition for Exams

Attribute:

An attribute is a property or characteristic of an entity in a database.

One-Mark Important Points

- Attributes are columns in a table.
- Attributes describe entities.
- Key attribute uniquely identifies records.
- Composite attribute can be divided.
- Multi-valued attribute stores multiple values.
- Derived attribute is calculated from another attribute.